COM3502/4502/6502
SPEECH PROCESSING

Lecture 15

Acoustic Phonetics

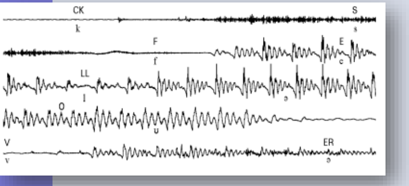
Articulatory Phonetics

… a description of speech sounds in terms of the *physical actions* performed in their production (*Lecture #14*)

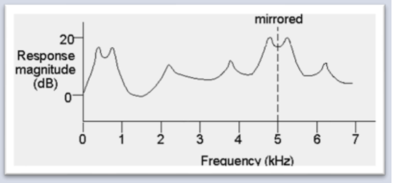
Acoustic Phonetics

… a description of speech sounds in terms of the *acoustic consequences* of their production (this lecture)

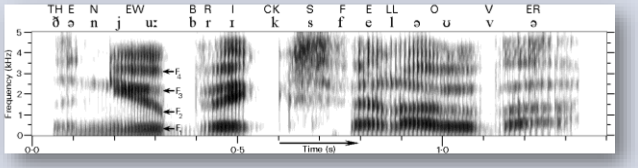
Acoustic Consequences



Waveforms



Spectra

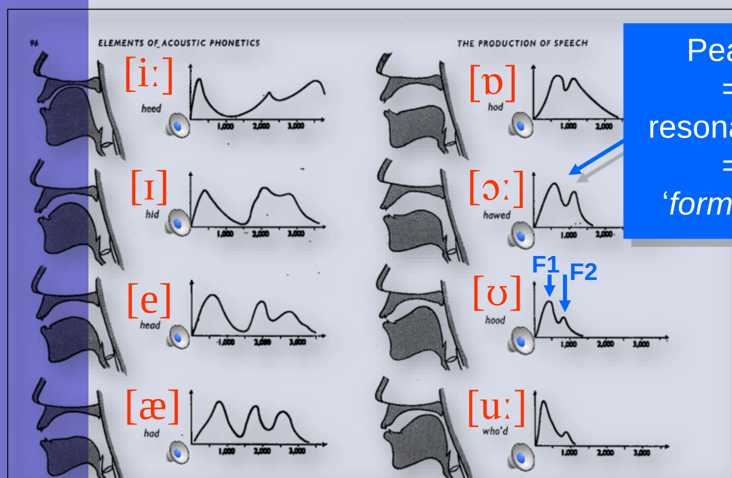


Spectrograms

Stationary Sounds

- Some speech sounds ('continuants') can be sustained over time and retain their phonetic quality
- For example …
  - monophthong vowels, e.g. [iː æ uː]
  - all fricatives, e.g. [f θ v ð s ʃ z ʒ]
  - some approximants, e.g. [ɹ l]
  - all nasals, e.g. [n m ŋ]

Vowel Spectra



Peaks
=
resonances
=
'*formants*'

Taken from: Ladefoged, P. (1962). *Elements of Acoustic Phonetics*: London: University of Chicago Press.

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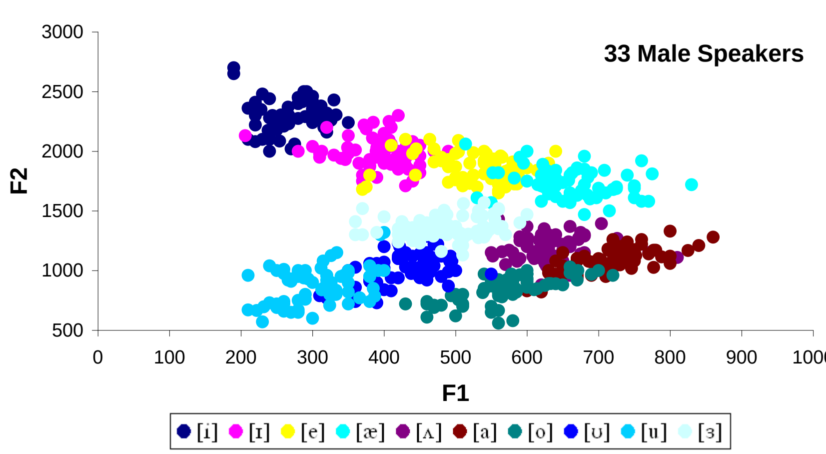
Peterson & Barney Vowel Data



Peterson, G. E., & Barney, H. L. (1952). Control methods used in a study of the vowels. *Journal of the Acoustical Society of America*, 24, 175-184.

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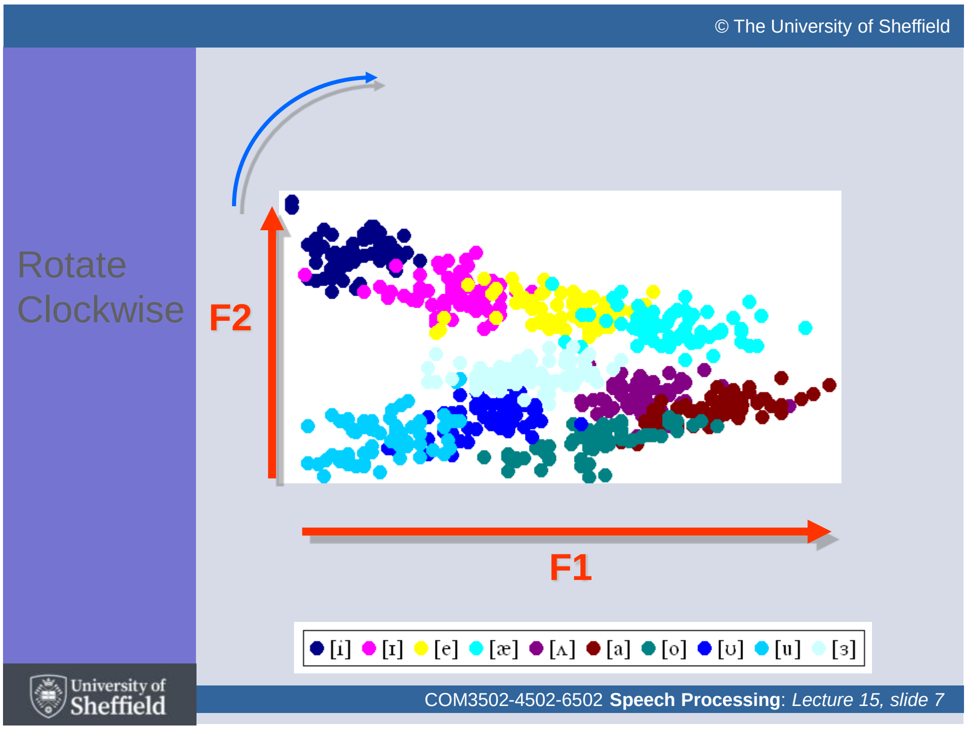
© The University of Sheffield
Rotate Clockwise
F2
F1
[i] [ɪ] [e] [æ] [ʌ] [ɑ] [o] [ʊ] [u] [ɜ]
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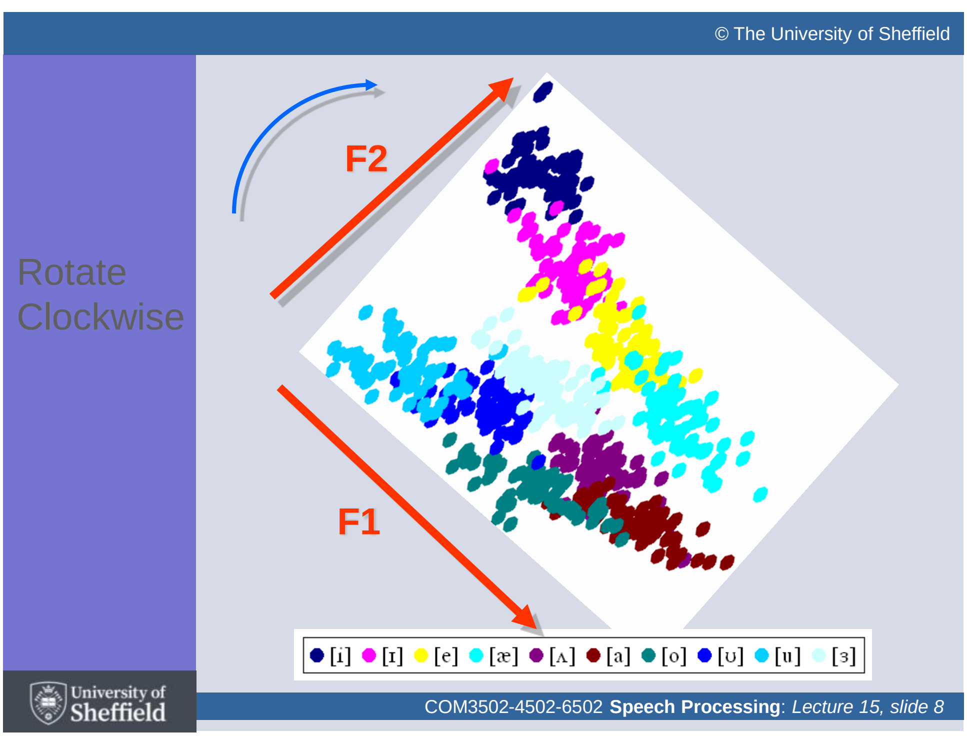
© The University of Sheffield
Rotate Clockwise
F2
F1
[i] [ɪ] [e] [æ] [ʌ] [ɑ] [o] [ʊ] [u] [ɜ]
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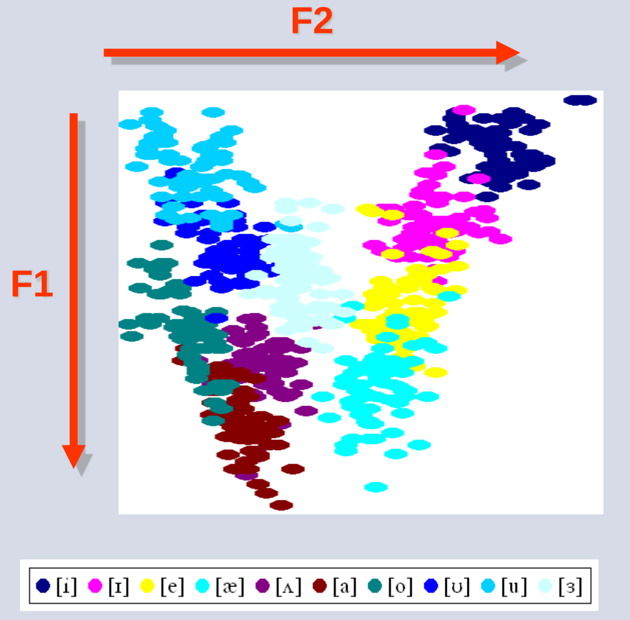
Reverse Horizontal Axis



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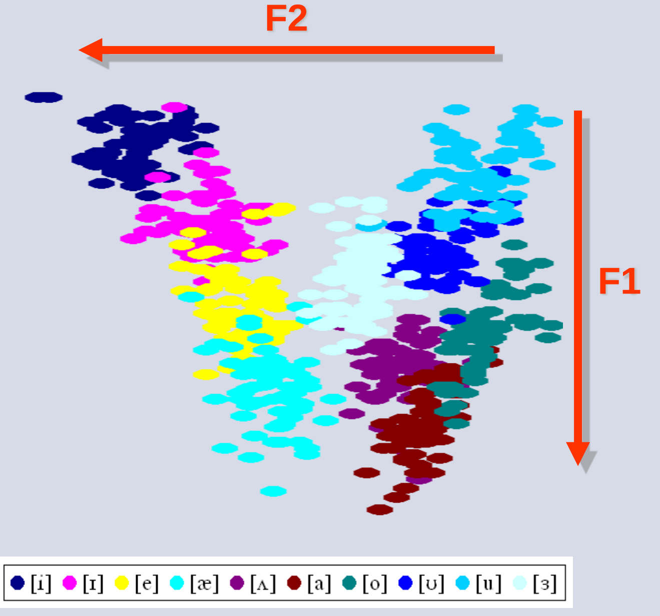
Reverse Horizontal Axis



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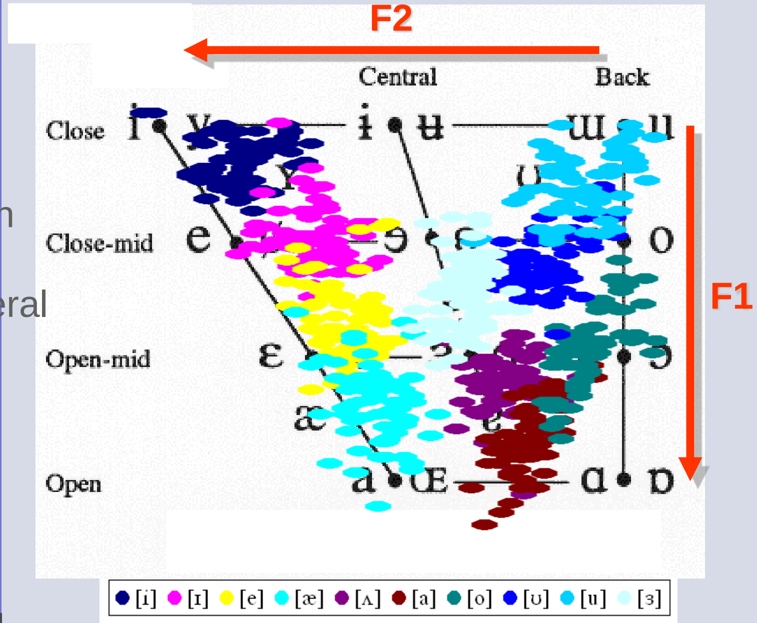
Overlay on the Vowel Quadrilateral



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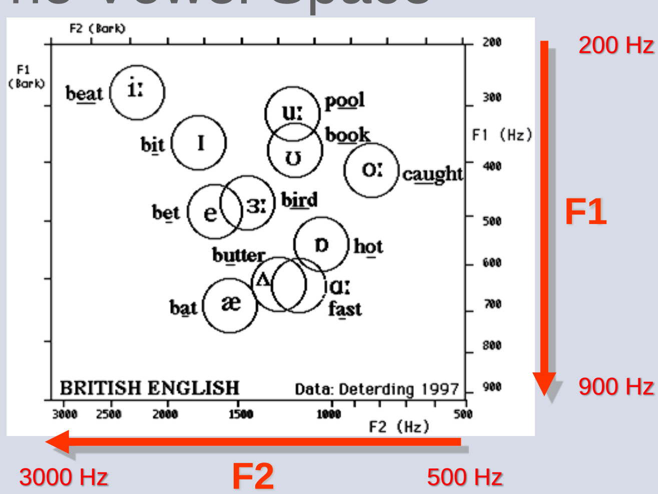
The Vowel Space



Taken from: Deterding, D. (1997). The formants of monophthong vowels in Standard Southern British English pronunciation. *Journal of the International Phonetic Association*, 27.

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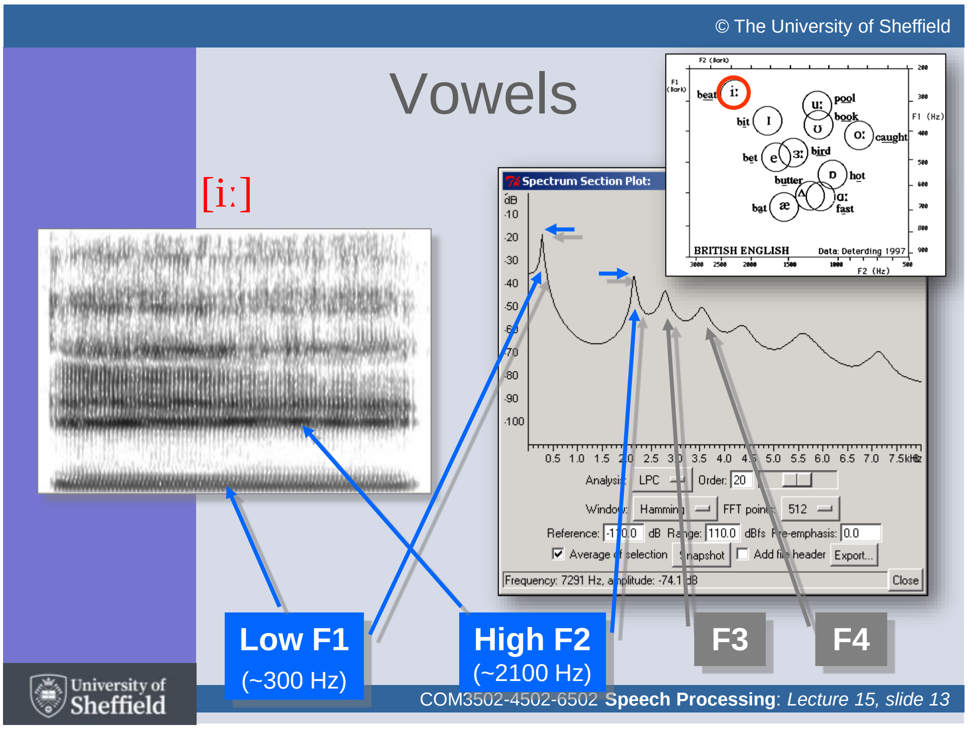

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Vowels
[iː]
Spectrum Section Plot:
BRITISH ENGLISH
Data: Deterding 1997
beat
bit
bet
butter
bat
pool
book
caught
bird
hot
fast
Low F1
(~300 Hz)
High F2
(~2100 Hz)
F3
F4
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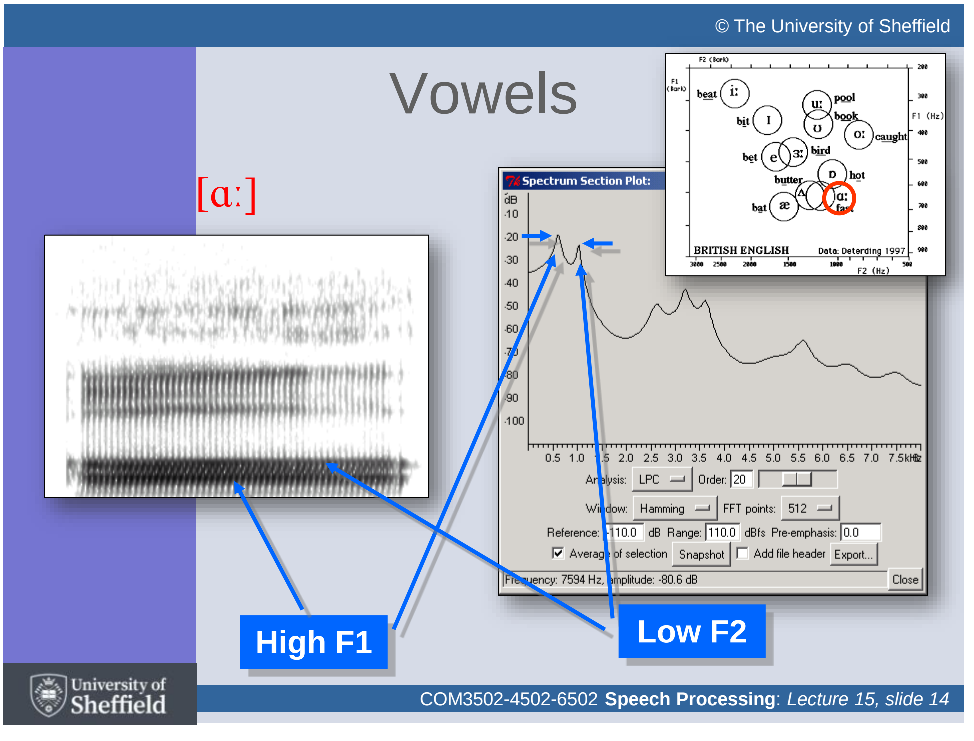

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Vowels
[ɑː]
Spectrum Section Plot:
BRITISH ENGLISH
Data: Deterding 1997
beat
bit
bet
butter
bat
pool
book
caught
bird
hot
fast
High F1
Low F2
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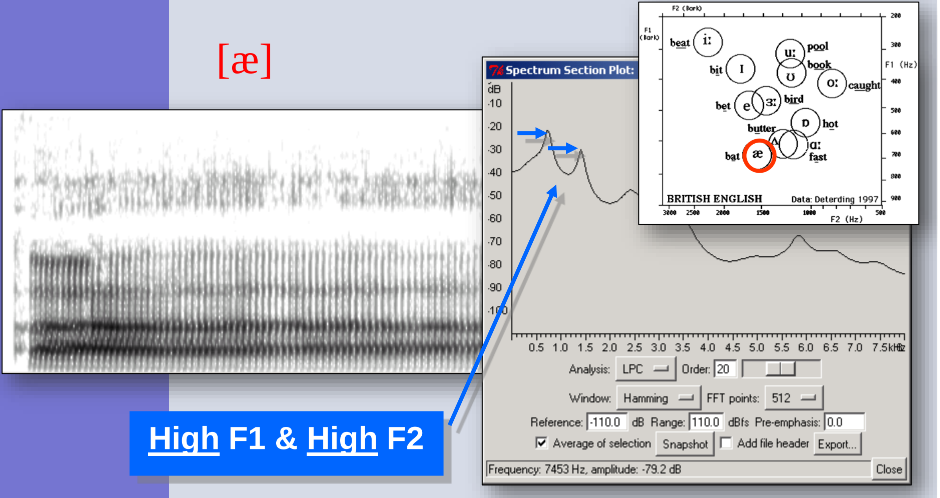
Vowels

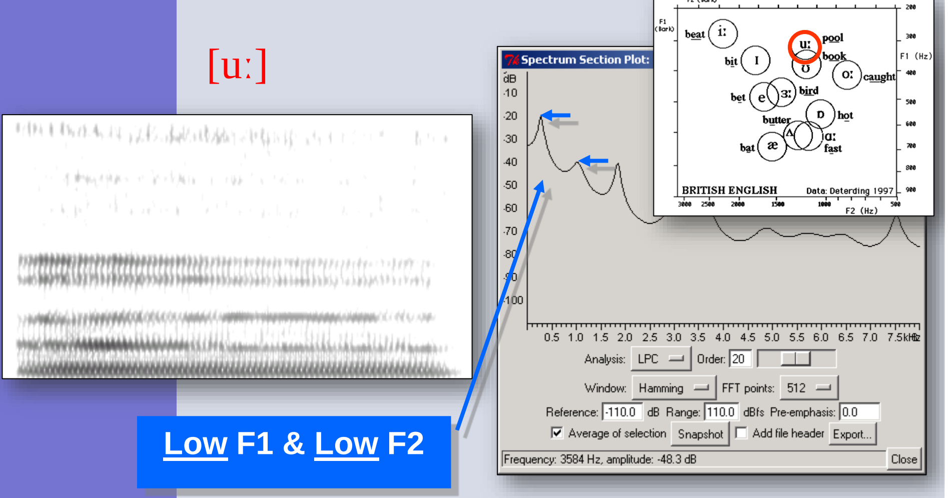
[æ]



High F1 & High F2

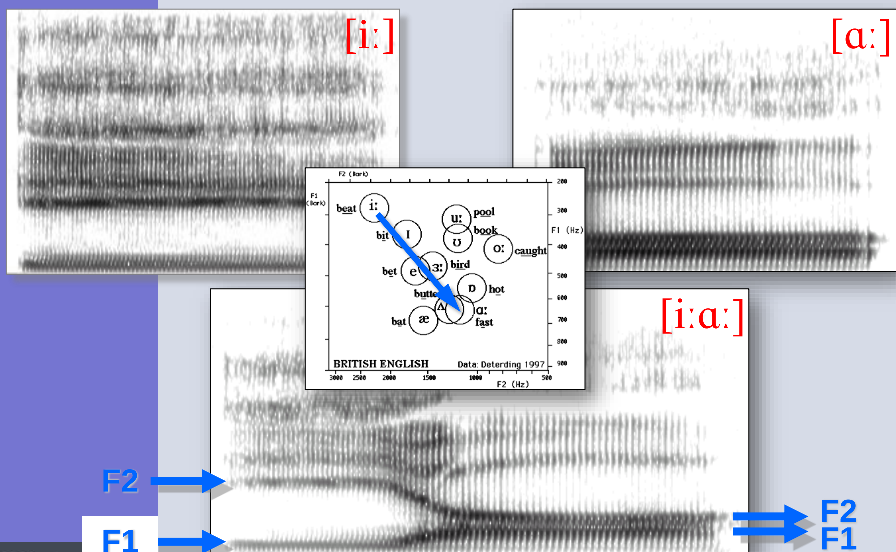
Vowels

[uː]



Low F1 & Low F2

Two Vowels



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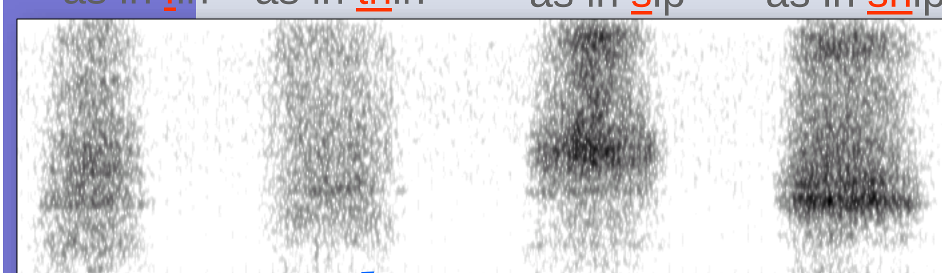
Voiceless Fricatives

[f] as in fin

[θ] as in thin

[s] as in sip

[ʃ] as in ship



No appreciable energy at low frequencies

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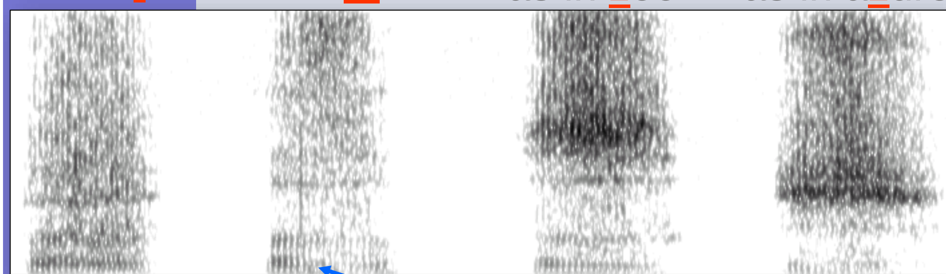
Voiced Fricatives

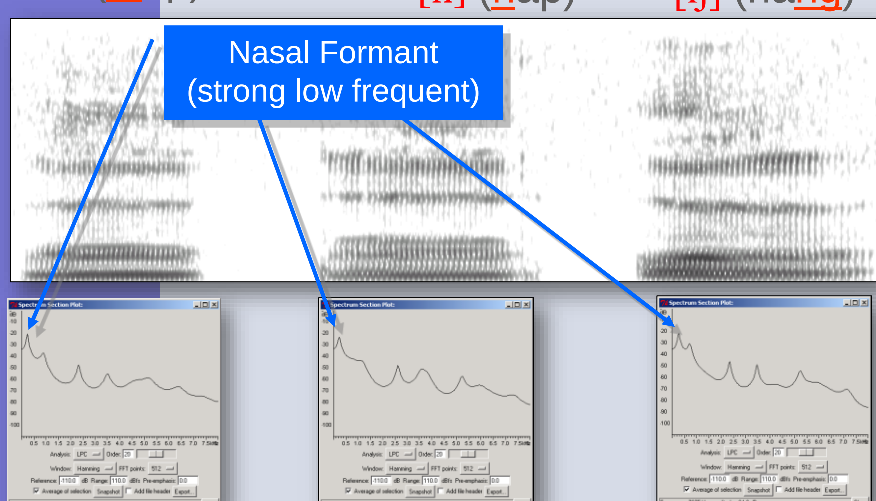
[v] as in fin | [ð] as in that | [z] as in zoo | [ʒ] as in azure



Voicing energy at low frequencies

Nasals

[m] (map) | [n] (nap) | [ŋ] (hang)



Non-Stationary Sounds

- The essence of some speech sounds is that they change over time
- For example …
  - diphthongs, e.g. [eɪ əʊ]
  - plosives, e.g. [p b t d k g]
  - affricates, e.g. [tʃ dʒ] (chin, jam)
  - glides, e.g. [w j] (wet, yet)

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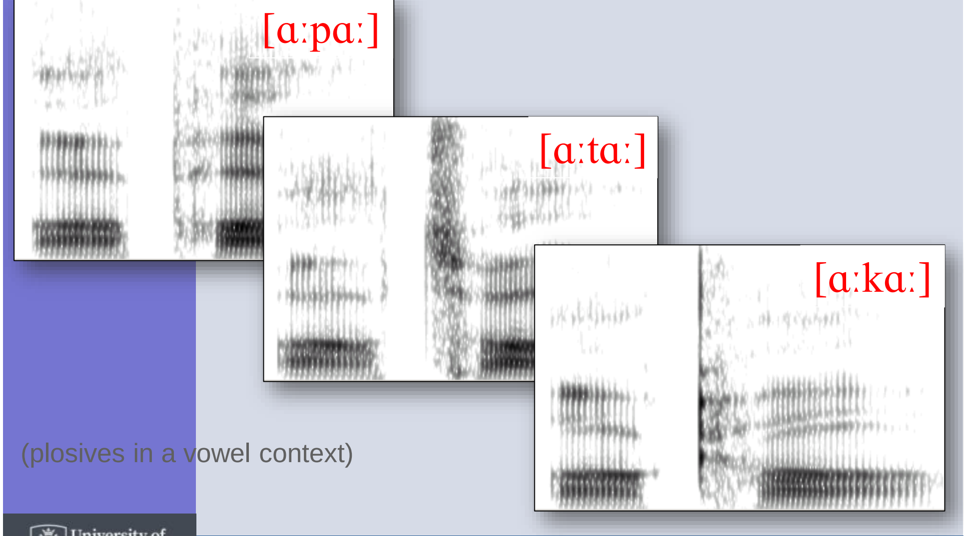
Voiceless Plosives



(plosives in a vowel context)

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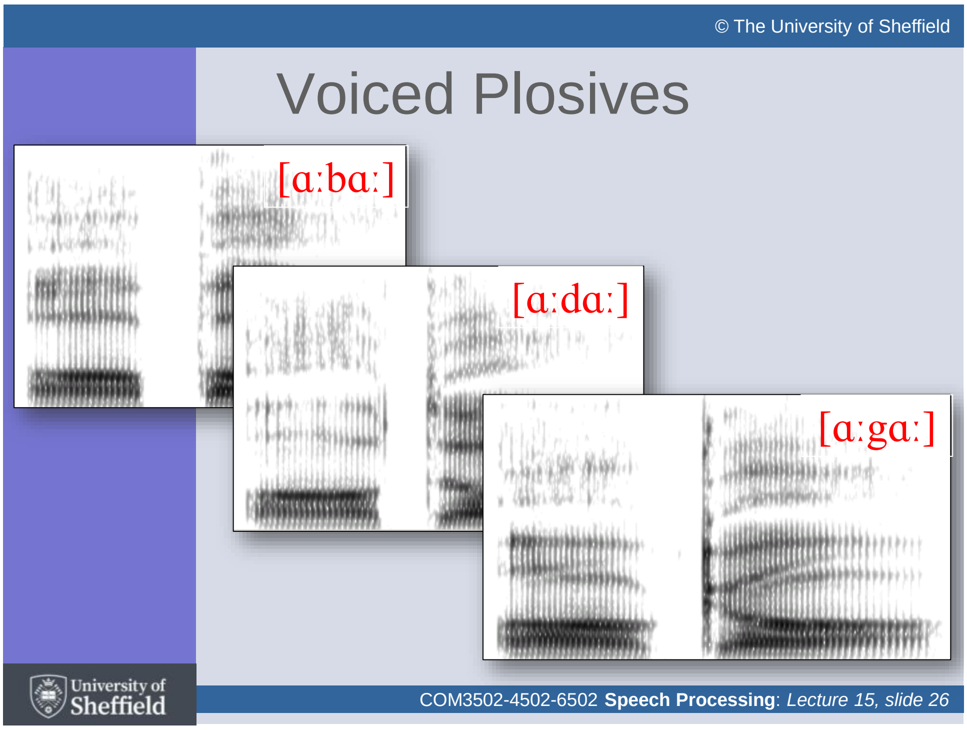

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Voiced Plosives
[ɑːbɑː]
[ɑːdɑː]
[ɑːgɑː]
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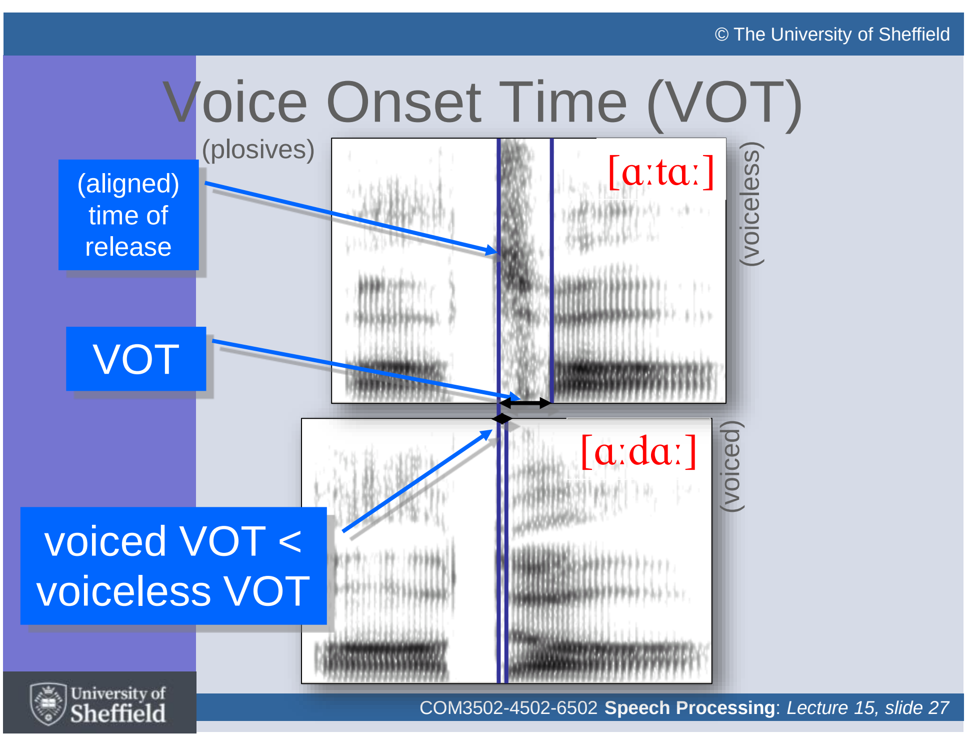

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Voice Onset Time (VOT)
(plosives)
(aligned) time of release
VOT
voiced VOT < voiceless VOT
[ɑːtɑː]
(voiceless)
[ɑːdɑː]
(voiced)
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COM3502-4502-6502 **Speech Processing**: *Lecture 15, slide 27*

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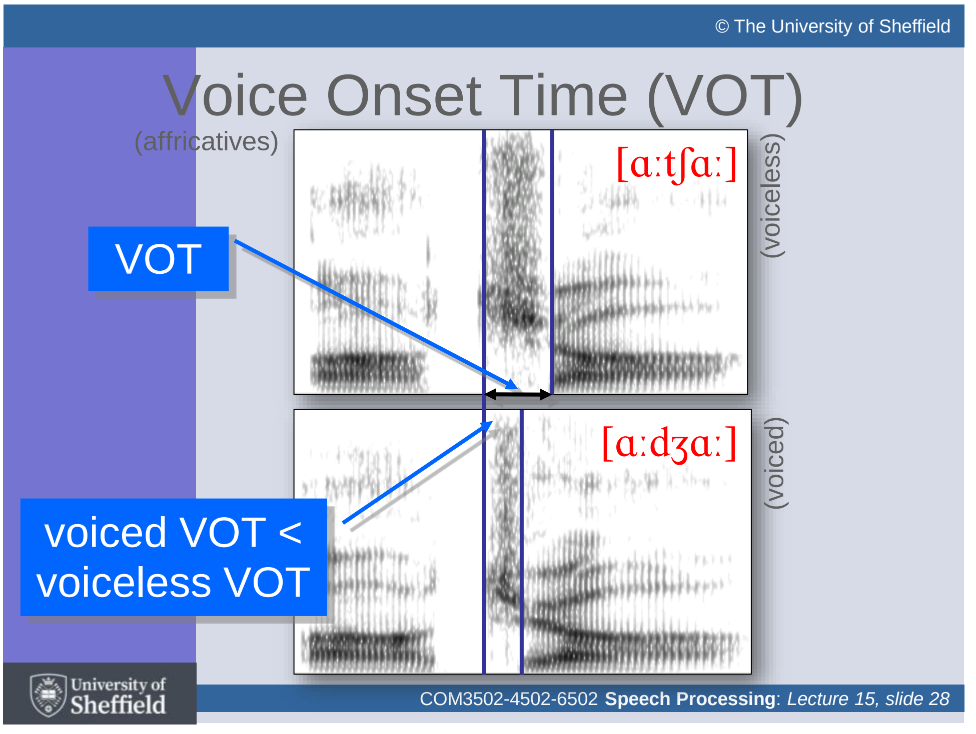
© The University of Sheffield
Voice Onset Time (VOT)
(affricatives)
VOT
[ɑːtʃɑː]
(voiceless)
[ɑːdʒɑː]
(voiced)
voiced VOT < voiceless VOT
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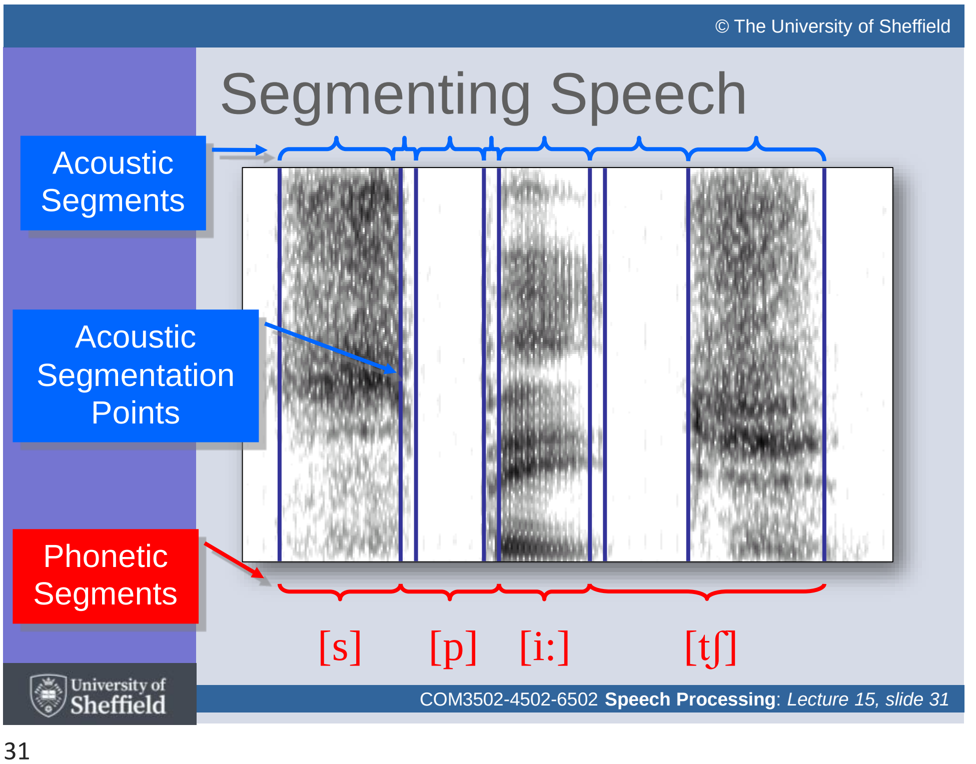
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Segmenting Speech
Acoustic Segments
Acoustic Segmentation Points
Phonetic Segments
[s]
[p]
[iː]
[tʃ]
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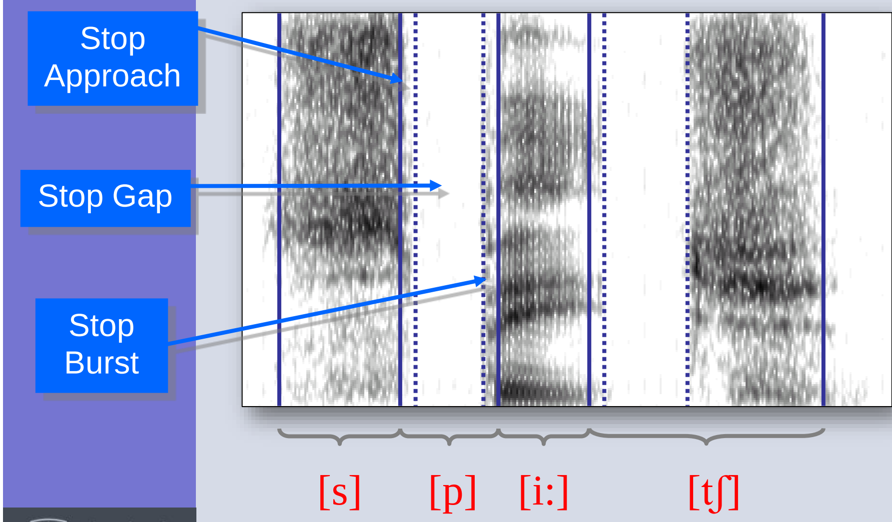
Segmenting Speech

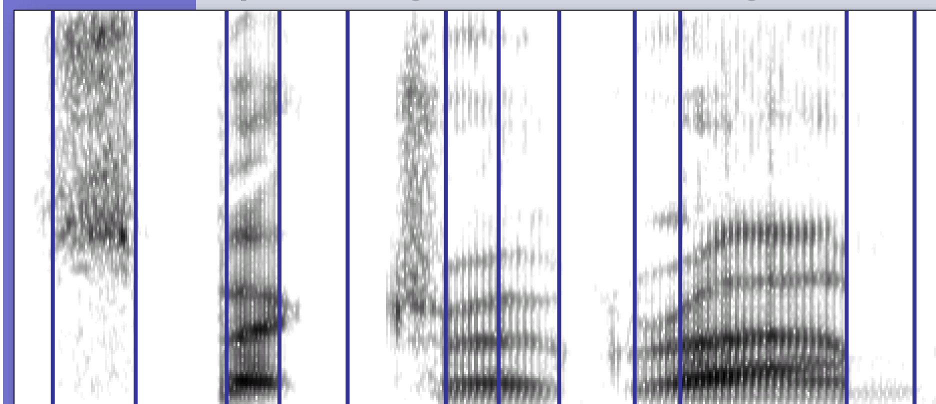


Spectrogram Reading

Victor Zue

- In the late 1970s, there was a general consensus (*among the technologists*) that there was insufficient information in the speech signal to perform automatic speech recognition
- In the 1980s, Victor Zue (MIT) taught himself (*and others*) to 'read' spectrograms, and proved everyone wrong

Spectrogram Reading



[s] [p] [e] [k] [t] [ɹ] [ə] [g] [ɹ] [æ] [m]

"spectrogram"

Coarticulation

{[p]:[s]_[i:]}

- 'Coarticulation' refers to the influence of one sound on another
- To a speech technologist/engineer, this is a form of 'context-dependency'
- To a phonetician, it is a consequence of efficient motor planning
- Coarticulations can be both short and long range (*e.g. the influence of nasalisation or lip rounding can span several syllables*)

Coarticulation

[w aɪ ɑː j uː ɜː l iː j uː aʊ l]

This slide set has covered …

- F1-F2 vowel space
- Fricatives
- Stops
- Voice onset time
- Close phonetic transcription
- Segmenting speech
- Whispering
- Coarticulation

Any Questions ?

Next slide set …

Phonology and Prosody